

Root-Protograph-Based Anytime Codes for Non-Ergodic Block-Fading Channels

Xiaoxi Yu[✉], Zhaojie Yang[✉], and Yong Liang Guan[✉], *Senior Member, IEEE*

Abstract—The study of anytime codes over non-ergodic block-fading (BF) channels remains largely unexplored. To address this gap, we propose a novel root-protograph-based anytime (RP-anytime) coding framework and develop modified density evolution (DE) equations. Leveraging these equations, we introduce two performance metrics to evaluate the asymptotic behavior of RP-anytime codes under the assumption of infinite frame length: the average asymptotic bit error rate (BER), which characterizes anytime reliability, and the anytime outage region, which reflects the code's ability to achieve full diversity. Using these metrics, we analyze the performance of the proposed RP-anytime codes and compare them with both root-protograph LDPC codes and conventional anytime codes. The results demonstrate the superiority of the RP-anytime structure in terms of reliability and diversity over non-ergodic BF channels. Furthermore, we employ a multi-objective differential evolution (MODE) algorithm to design optimized RP-anytime codes. Simulation results confirm that the newly designed codes achieve excellent anytime reliability, full diversity, and near-outage-limit error performance—significantly outperforming prior-art root-protograph LDPC codes and anytime codes.

Index Terms—Anytime codes, non-ergodic block-fading (BF) channel, density evolution, protograph low-density parity-check (LDPC) codes, multi-objective differential evolution (MODE).

I. INTRODUCTION

ANYTIME information theory plays a crucial role in managing unstable processes and controlling unstable systems over noisy channels [1]. Unlike traditional information theory, which focuses on achieving error-free decoding with potentially infinite delay, anytime information theory requires that the probability of decoding errors decreases continuously and exponentially as the decoding delay (i.e., decoding window length) increases [2]. Codes that satisfy this requirement are referred to as *anytime codes*, characterized by their *anytime-reliable property* [3]. This type of coding scheme is particularly useful in communication scenarios where data must be transmitted quickly but can initially tolerate some

inaccuracies, with the expectation that errors will be refined or corrected over time. Typical applications include real-time video streaming [4], autonomous vehicles [5], and Internet of Things (IoT) devices [6].

To date, practical anytime codes have been developed for ergodic channels, such as the binary erasure channel (BEC) and the binary-input additive white Gaussian noise (AWGN) channel. A class of low-density parity-check (LDPC) convolutional codes was shown to exhibit anytime-reliable property over the BEC [7], which was later extended to the AWGN channel [3]. A subtype of spatially coupled LDPC (SC-LDPC) codes demonstrated anytime reliability over both BEC and BI-AWGN channels [8], offering improved decoding performance with low encoding complexity. Recent studies have further enhanced the decoding capabilities of SC-LDPC anytime codes [9], [10] and introduced protograph-based anytime (P-anytime) codes to improve finite-length decoding performance [11]. These developments have broadened the applicability of anytime codes, including their use in relay channels [12], chaotic systems [13], joint source-channel coding [14], and rate-compatible coding for incremental redundancy hybrid automatic repeat request (IR-HARQ) schemes [15].

However, to the best of our knowledge, the performance of anytime codes over non-ergodic block-fading (BF) channels has not yet been investigated, and no existing anytime code is specifically designed for such channels. The non-ergodic BF channel is a practical model that describes wireless scenarios where fading varies slowly [16]. In a BF channel, multiple symbols experience the same fading coefficient, meaning that each codeword is affected by several fading gains modeled as independent and identically distributed (i.i.d.) random variables. Representative applications include multicarrier modulation using orthogonal frequency division multiplexing (OFDM) and slow time-frequency hopping systems [17], [21].

Due to their non-ergodic nature, BF channels present different challenges for error-correction code (ECC) design compared to ergodic channels (e.g., fast-fading channels). In a BF channel, the Shannon capacity is zero because there is an irreducible probability of decoding failure. As a result, outage probability is used as a fundamental performance metric to characterize the lowest achievable word error rate (WER) for any coding scheme [16]. For instance, LDPC codes that approach capacity in AWGN channels do not approach the outage limit in BF channels. Furthermore, the diversity order is defined to quantify the slope of the WER curve, and codes

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Xiaoxi Yu and Yong Liang Guan are with the School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore 639798 (e-mail: xiaoxiyu@ntu.edu.sg; eylguan@ntu.edu.sg).

Zhaojie Yang is with the School of Information Engineering, Guangdong University of Technology, Guangzhou 510006, China (e-mail: yangzhaojie@gdut.edu.cn).

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that achieve a diversity order equal to the number of fading gains per codeword are referred to as full diversity codes.

In [17], the authors proposed a family of structured LDPC code ensembles, known as root-LDPC codes, specifically designed for non-ergodic BF channels. These codes achieve full diversity and exhibit decoding performance close to the outage limit under iterative decoding. Root-protograph-based LDPC codes were later developed in [18], offering lower encoding complexity while preserving full diversity and near-outage-limit performance. More recently, [19] and [20] investigated SC-LDPC codes for BF channels, showing improved diversity compared to root-LDPC codes. However, these designs are not well-suited for anytime-coded transmissions. This limitation motivates the development of novel anytime codes that provide strong anytime reliability while also achieving full diversity in non-ergodic BF channels.

Therefore, this paper addresses the above gap by making the following contributions:

- We propose a novel root-protograph-based anytime (RP-anytime) coding structure and derive the corresponding modified density evolution (DE) equations for performance analysis under the anytime coding framework.
- Based on the modified DE analysis, we introduce two asymptotic performance metrics: (i) the average asymptotic bit error rate (BER), which characterizes the anytime reliability of the code, and (ii) the anytime outage region, which reflects the code's ability to achieve full diversity, over non-ergodic BF channels.
- We evaluate the asymptotic decoding performance of the proposed RP-anytime codes using the two metrics. The results demonstrate that the proposed structure achieves improved anytime reliability compared to root-protograph-based LDPC codes, and full diversity compared to conventional anytime codes. These analytical results are further validated by finite-length simulations.
- Finally, we employ the proposed performance metrics as optimization objectives within a multi-objective differential evolution (MODE) algorithm to design RP-anytime codes for code rates $R=1/2$ and $R=1/3$. Simulation results show that the optimized codes outperform randomly constructed RP-anytime codes, offering enhanced anytime reliability and error performance that approaches the theoretical outage limit.

II. SYSTEM MODELS

A. Anytime-Coded Transmission

The fundamental distinction between anytime-coded transmission and conventional ECC transmission lies in their design principles for both encoding and decoding. Conventional ECCs aim to ensure reliability at the entire codeword level, employing fixed-length codewords and deferred decoding (i.e., decoding occurs only after the entire codeword has been received). In contrast, anytime-coded systems are designed to provide progressively improving reliability at a finer frame level, where the error probability of each frame decreases

continuously as the decoding delay (i.e., decoding window length) increases [1], [2].

1) *Encoding of Anytime Codes*: In anytime-coded transmission, the source generates a K -bit information frame $\mathbf{m}_p \in \{0, 1\}^K$ at each discrete time index $p \geq 1$. Each newly generated information frame is received and stored by the anytime encoder. Upon receiving $\mathbf{m}_t$, $t \geq p \geq 1$, the encoder re-encodes all information frames collected up to time index t , denoted as $\mathbf{m}_{[1,t]} = [\mathbf{m}_1, \mathbf{m}_2, \dots, \mathbf{m}_p, \dots, \mathbf{m}_t]$, into a fixed-length encoded frame $\mathbf{c}_t \in \{0, 1\}^N$.

Although the length of the input sequence $\mathbf{m}_{[1,t]}$ increases with time, the output $\mathbf{c}_t$ always remains a fixed length of N bits. In other words, an increasing amount of information is compactly represented within a fixed-size frame.

The encoded frame $\mathbf{c}_t$ is then modulated into the transmitted frame $\mathbf{x}_t$ for transmission over the channel at time index t .¹ For simplicity, we will refer to $\mathbf{x}_t$ directly as the encoded frame throughout the remainder of the paper.

As illustrated in Fig. 1(a), each information frame $\mathbf{m}_p$ contributes not only to the current encoded frame $\mathbf{x}_p$, but also to all subsequent transmissions $\mathbf{x}_{p+1}, \mathbf{x}_{p+2}, \dots, \mathbf{x}_t$. Thus, each encoded frame $\mathbf{x}_t$ carries the cumulative information of all information frames up to time index t . Despite this growing dependency, the size of each encoded frame remains fixed at N bits. This causal mapping can be realized using structures such as syndrome formers [22], resulting in a parity-check matrix with a lower-triangular structure whose dimensions grow with t .

2) *Decoding of Anytime Codes*: At the receiver, a buffer is used to store the most recently decoded frames, which are combined with each newly received frame for subsequent decoding.

At time $t=1$, the encoded frame $\mathbf{x}_1$ is transmitted, received, and demodulated as $\mathbf{y}_1$. The anytime decoder uses the corresponding portion of the parity-check matrix (with dimension $M \times N$) to decode $\mathbf{y}_1$ and produce the decoded frame $\hat{\mathbf{x}}_1(1)$, where the number in parentheses indicates the decoding time index. The decoded frame $\hat{\mathbf{x}}_1(1)$ is then stored in the buffer for future decoding.

At $t=2$, the newly received frame $\mathbf{y}_2$ is input to the decoder, which jointly processes $\hat{\mathbf{x}}_1(1)$ and $\mathbf{y}_2$ using an expanded parity-check matrix of dimension $2M \times 2N$ (i.e., expanding-window decoding). This results in a new set of decoded frames: $[\hat{\mathbf{x}}_1(2), \hat{\mathbf{x}}_2(2)]$. These updated frames are stored in the buffer, and the earlier estimate $\hat{\mathbf{x}}_1(1)$ is discarded.

This process continues iteratively. At time $t=p$, the received frame $\mathbf{y}_p$ is decoded jointly with the buffered $[\hat{\mathbf{x}}_1(p-1), \hat{\mathbf{x}}_2(p-1), \dots, \hat{\mathbf{x}}_{p-1}(p-1)]$ using a parity-check matrix of size $pM \times pN$. The decoder then produces an updated set of estimates $[\hat{\mathbf{x}}_1(p), \hat{\mathbf{x}}_2(p), \dots, \hat{\mathbf{x}}_{p-1}(p), \hat{\mathbf{x}}_p(p)]$, which replace the previous buffered frames and are stored for use in the next decoding step.

Since we already know that each encoded frame $\mathbf{x}_p$ (and consequently $\mathbf{y}_p$) carries the cumulative information of all information frames $\mathbf{m}_{[1,p]}$, the decoded frame $\hat{\mathbf{x}}_1(2)$ will

¹This paper considers binary phase-shift keying (BPSK) modulation. However, the proposed framework can be extended to support higher-order modulations to improve spectral efficiency.

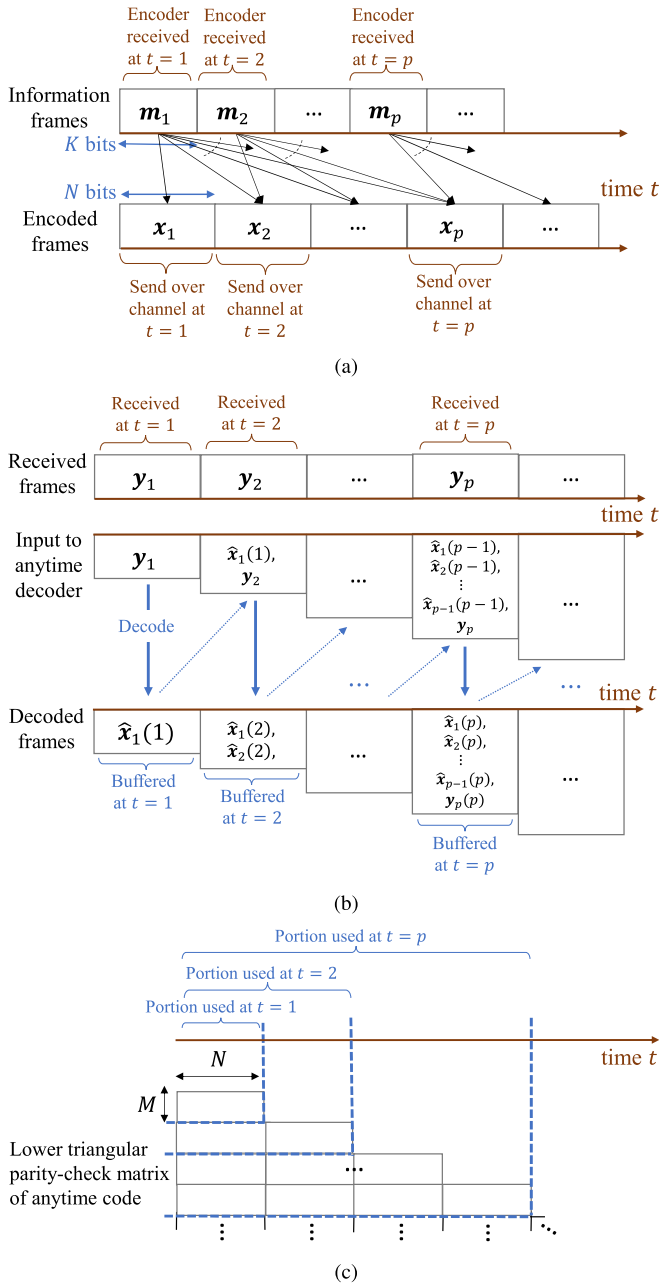


Fig. 1. (a) Encoding process; (b) Decoding process; (c) Parity-check matrix used in expanding-window decoding of the anytime codes.

contain fewer errors than $\hat{x}_1(1)$. This is because at $t=2$, y_2 is decoded jointly with $\hat{x}_1(1)$, thereby providing additional information about x_1 . As this process continues, the bit error rate (BER) of $\hat{x}_1(t)$ decreases monotonically with increasing t .

To formally describe this behavior, we introduce the term *decoding delay* d , which refers to the difference between the current decoding time and the original transmission time of a frame. For example, when $t=p$, the decoded frame $\hat{x}_1(p)$ has a decoding delay of $d=p-1$, $\hat{x}_2(p)$ has $d=p-2$, $\hat{x}_{p-1}(p)$ has $d=1$, and $\hat{x}_p(p)$ has $d=0$. In other words, the BER of each frame improves as its decoding delay d increases.

This decoding process is illustrated in Fig. 1(b), while Fig. 1(c) shows the corresponding parity-check matrix used

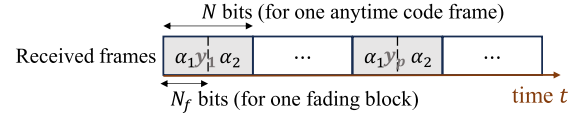


Fig. 2. Illustration of non-ergodic BF channels with $F = 2$ for anytime codes. The length of each frame is N bits.

at different decoding time indices t , providing a clear illustration of the expanding-window message-passing decoding procedure [3].

3) *Anytime-Reliable Property*: At each time index p , after obtaining the decoded frames, the estimated information frames $[\hat{m}_1(p), \hat{m}_2(p), \dots, \hat{m}_p(p)]$ can also be derived. Therefore, if we focus on the bit error rate (BER) performance of a single frame m_p as its decoding delay d increases, the anytime-reliable property of a given code transmitted over an arbitrary channel is defined by the following relationship [23]:

$$P_e(p, d) = \Pr(\hat{m}_p(p+d) \neq m_p \mid x_{[1,p+d]} \text{ was transmitted}) \leq \beta e^{-\varepsilon d}, \quad (1)$$

where β is a constant determined by the channel parameters, and ε , known as the *delay-exponent*, characterizes the exponential rate at which the BER $P_e(p, d)$ decreases with increasing d . Specifically, $-\varepsilon$ represents the slope of the $P_e(p, d)$ curve when plotted against d on a semi-logarithmic scale.

As a result, if a delay-exponent $\varepsilon \geq 0$ exists, the BER $P_e(p, d)$ will decay to an arbitrarily small value as the decoding delay d becomes sufficiently large. A code that satisfies this property is referred to as an *anytime code*.

B. Non-Ergodic Block Fading Channels

1) *Channel Model*: In this paper, we analyze the non-ergodic BF channel environment for anytime-coded transmission. Each encoded frame x_p transmitted at time index p is subject to F ($F \geq 2$) independent fading gains. Accordingly, x_p is partitioned into F fading blocks, each of length $N_f = N/F$ bits, where each block is affected by a distinct fading gain α_f , for $f = 1, \dots, F$. The j^{th} symbol of the received frame y_p at time p is given by:

$$y_{p,j} = \alpha_f x_{p,j} + z_{p,j}, \quad (2)$$

where $y_{p,j} \in \mathbb{R}$, $p = 1, \dots, t$, $j = 1, \dots, N$, and $f = 1 + \lfloor (j-1)/N_f \rfloor$, with $\lfloor \cdot \rfloor$ denoting the floor function. The additive noise terms $z_{p,j}$ are i.i.d. Gaussian variables, $z_{p,j} \sim \mathcal{N}(0, \sigma_n^2)$, with variance $\sigma_n^2 = N_0/2$, where N_0 is the one-sided noise power spectral density. The fading coefficients α_f are modeled as i.i.d. Rayleigh random variables with $E[\alpha_f^2] = 1$.

Fig. 2 illustrates an example of anytime-coded transmission over a non-ergodic BF channel with two fading gains, where the first and second halves of each frame are affected by α_1 and α_2 , respectively.

For a given code rate R , and assuming perfect channel state information (CSI) at the receiver, the instantaneous received

signal-to-noise ratio (SNR) per symbol for the f^{th} fading block is given by [18]:

$$\gamma_f = \frac{E_s}{N_0} \alpha_f^2 = \frac{E_b}{N_0} R \alpha_f^2, \quad (3)$$

where E_s and E_b denote the average energy per symbol and per bit, respectively. Consequently, we define the received SNR vector as $\gamma = (\gamma_1, \dots, \gamma_F)$, and the fading vector as $\alpha = (\alpha_1, \dots, \alpha_F)$.

In non-ergodic BF channels, the channel capacity is significantly influenced by the fading coefficients α_f , rendering the channel information-unstable [24]. Due to this non-ergodicity, the Shannon capacity is effectively zero. Therefore, instead of using capacity as a performance metric, the reliability of BF channels is typically evaluated in terms of outage probability, which serves as a fundamental lower bound on the WER for any coding strategy. The outage probability is defined as [16]:

$$P_{\text{out}} = \mathbb{P}(I(\mathbf{x}; \mathbf{y} | \alpha) < R), \quad (4)$$

where $I(\cdot)$ denotes the instantaneous mutual information between the channel input and output. An outage event occurs when the mutual information, conditioned on the fading realization α , falls below the target code rate R .

2) *Anytime Coding for Non-Ergodic BF Channel*: Effective code design for non-ergodic BF channels must exploit the specific properties of the channel. A key characteristic is that each fading block within a frame is affected by an independent fading coefficient, which introduces diversity. To mitigate the risk of all symbols being simultaneously corrupted by deep fades, the F fading blocks in a frame are jointly encoded.

At high SNR levels, the diversity order of a code—which determines the slope of the frame error rate (FER) curve on a log-log scale—is defined as [25]:

$$d_v = - \lim_{\gamma \rightarrow \infty} \frac{\log(P_{\text{fe}})}{\log(\gamma)}, \quad (5)$$

where P_{fe} denotes the frame error rate (FER) at the decoder output and γ represents the received SNR. Note that FER is used here for anytime codes, defined as a frame-level error metric, obtained by aggregating the information bit error rates within each frame, as opposed to the WER commonly used in conventional whole-codeword coding.

The maximum achievable diversity order is constrained by a Singleton-like bound [26]:

$$d_v \leq 1 + \lfloor F(1 - R) \rfloor, \quad (6)$$

where R is the code rate. Therefore, for a code with rate $R = 1/F$, the maximum possible diversity order is F . A code is said to achieve full diversity if it satisfies $d_v = F$ [25], [27].

III. PROPOSED RP-ANYTIME CODES AND MODIFIED DENSITY EVOLUTION

A. Design and Construction of RP-Anytime Codes

An RP-anytime code ensemble is constructed by spatially coupling a sequence of independent root-protographs. Each root-protograph is a small bipartite graph comprising b_v types of variable nodes (VNs) and b_c types of check nodes (CNs),

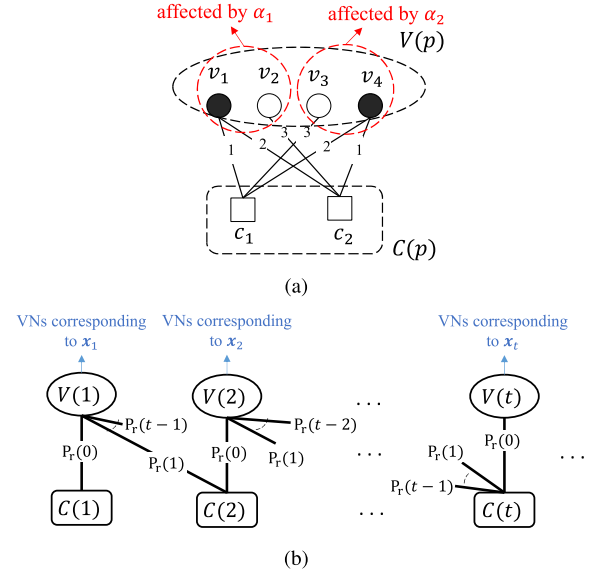


Fig. 3. (a) A standard root-protograph. (b) The protograph of the RP-anytime codes. $\text{Pr}(h)$ represents the probability of an edge connecting two root-protographs following (7).

where the CNs include F distinct root-checks to provide protection against block fading. An example root-protograph, denoted as $\mathbf{B}_{\text{Reg-RP}}$ [18], is shown in Fig. 3(a). In this graph, black circles represent two types of information VNs, white circles represent two types of parity VNs, and squares denote two types of CNs.

In this example, VNs v_1 and v_2 are subject to the first fading gain α_1 , while v_3 and v_4 are affected by the second fading gain α_2 . The two root-checks, c_1 and c_2 , are designed to protect the information VNs associated with α_1 and α_2 , respectively, thereby ensuring reliability across distinct fading conditions.

For the p^{th} root-protograph—i.e., the protograph instance used to generate the code frame transmitted at time $t = p$ —we denote the sets of variable and check nodes as $V(p)$ and $C(p)$, respectively. After lifting, the variable node set $V(p)$ corresponds to the encoded frame $\mathbf{x}_p$, which is transmitted over the channel at time index p . For simplicity, we will use the terms “ p^{th} root-protograph” and “VNs transmitted at time p ” interchangeably throughout the remainder of this paper.

To implement spatial coupling in the root-protographs, the VN set $V(p)$ of the p^{th} root-protograph is connected to the CN sets $C(q)$ in root-protographs indexed by $q \in \{p, \dots, p+w-1\}$, where the coupling width w theoretically approaches infinity.² These connections follow an exponential distribution [8], with the probability of a VNs of the p^{th} root-protograph connecting to a CN of the q^{th} root-protograph described by:

$$\text{Pr}(h) = \frac{e^{-h\lambda}(1 - e^{-\lambda})}{1 - e^{-w\lambda}} \xrightarrow{w \rightarrow \infty} e^{-h\lambda}(1 - e^{-\lambda}), \quad (7)$$

where $h = q - p$ denotes the relative position between connected VN and CN sets. The connection probabil-

²In theory, during transmission, message bits are encoded and decoded incrementally, causing w to grow linearly with time t . However, in practice, a large but finite value of w is adopted to enable tractable simulation and implementation.

ity $\Pr(h)$ decreases exponentially with increasing h , i.e., $\Pr(0) > \Pr(1) > \dots > \Pr(w-1)$. The parameter $\lambda \in (0, 1)$ is referred to as the exponential coupling rate, which controls the concentration of the distribution. Specifically, for a fixed h , a larger λ results in a higher $\Pr(h)$, leading to a more concentrated coupling around p . Based on prior works [10], [11], [15], setting $\lambda = 0.5$ offers a good trade-off between BER performance and decoding delay for the anytime code structure. Therefore, in all subsequent simulations and design algorithms, we fix the coupling rate to $\lambda = 0.5$.

Fig. 3(b) illustrates the structure of an RP-anytime code ensemble consisting of t root-protographs. It is important to note that each type of VN is only allowed to connect to its corresponding CN type, as defined in the base root-protograph. For example, node v_2 in $V(1)$ can only connect to c_2 in $C(1), C(2), \dots, C(t)$. Additionally, each VN has a fixed node degree, meaning that while the CN set it connects to may vary with time, the total number of edges (i.e., its degree) remains constant.

The convolutional base matrix associated with the constructed RP-anytime code ensemble at time t can be represented as:

$$\mathbf{B}_{[0,t-1]} = \begin{bmatrix} \mathbf{B}_0 & & & \\ \mathbf{B}_1 & \mathbf{B}_0 & & \\ \vdots & \vdots & \ddots & \\ \mathbf{B}_{t-1} & \mathbf{B}_{t-2} & \cdots & \mathbf{B}_0 \end{bmatrix}, \quad (8)$$

where each matrix $\mathbf{B}_h$ for $0 \leq h \leq t-1$ has dimensions $b_c \times b_v$. These matrices represent the coupling structure, indicating how the b_v VNs at position p connect to the b_c CNs at position $p+h$. This convolutional arrangement enables the code to accumulate memory over time, a key feature for achieving anytime reliability.

To construct the full parity-check matrix of the RP-anytime code, a *lifting* operation (i.e., copy-and-permute) is applied to the base matrix $\mathbf{B}_{[0,t-1]}$ using a lifting factor M_f . This operation produces the frame-structured parity-check matrix:

$$\mathbf{H}_{[0,t-1]} = \begin{bmatrix} \mathbf{H}_0 & & & \\ \mathbf{H}_1 & \mathbf{H}_0 & & \\ \vdots & \vdots & \ddots & \\ \mathbf{H}_{t-1} & \mathbf{H}_{t-2} & \cdots & \mathbf{H}_0 \end{bmatrix}, \quad (9)$$

where each frame $\mathbf{H}_k$ ($0 \leq k \leq t-1$) is of size $M \times N$, with $M = b_c M_f$ and $N = b_v M_f$.

The code rate at time t can then be computed as:

$$\begin{aligned} R &= \frac{Nt - Mt}{Nt} \\ &= \frac{b_v M_f t - b_c M_f t}{b_v M_f t} \\ &= \frac{b_v - b_c}{b_v}, \end{aligned} \quad (10)$$

under the assumption that the code is systematic and the parity-check matrix is full rank.

Note that, although the construction of the proposed RP-anytime codes follows the same exponentially distributed spatial coupling as prior anytime codes [8], [9], [11],

[15], these existing anytime code constructions are based on unstructured protographs. In contrast, the proposed RP-anytime codes are constructed from a root-protograph with predefined VN and CN types, and the anytime code construction process, including spatial coupling and lifting, is performed under explicit type and connection constraints inherited from the root-protograph.

B. Modified DE Equations

Unlike the DE analysis for conventional root-LDPC codes, which is codeword-based and characterizes the asymptotic error performance of a single codeword [17], the DE formulation for the proposed RP-anytime codes is frame-indexed, where the decoding delay is determined by the relative indices of the frame reception time t and the protograph position p . This delay-dependent DE analysis is necessary to characterize the anytime reliability and outage behavior of the proposed RP-anytime codes under BF channels.

To derive the DE equations for RP-anytime codes, we apply the Gaussian approximation technique from [28] to the underlying protograph structure of the RP-anytime code ensemble. It is important to note that, in DE analysis, the frame length is assumed to approach infinity, resulting in a cycle-free graph. Under this assumption, each symbol is directly associated with its corresponding VN in the base protograph matrix $\mathbf{B}$ of the RP-anytime codes.

In this method, the message exchanged between nodes is characterized by the mean of the log-likelihood ratio (LLR). The LLR for a received symbol y_j is given by:

$$\begin{aligned} L_{\text{ch},j} &= \ln \left(\frac{\Pr(x_j = +1 | y_j, \alpha_f)}{\Pr(x_j = -1 | y_j, \alpha_f)} \right) \\ &= \frac{2\alpha_f}{\sigma_n^2} y_j. \end{aligned} \quad (11)$$

Hence, the channel LLR follows a Gaussian distribution:

$$L_{\text{ch},j} \sim \mathcal{N} \left(\frac{2\alpha_f^2}{\sigma_n^2}, \frac{4\alpha_f^2}{\sigma_n^2} \right).$$

After receiving the set of frames $\mathbf{y}_{[1,t]}$, the following notations are used in the DE equations:

- $L_{v,t}^l(i, j, p)$: In the l^{th} iteration, this denotes the mean of the LLR message passed from VN v_j in the p^{th} protograph to a CN c_i in one of the $\{p, \dots, p+w-1\}$ protographs.
- $L_{c,t}^l(i, j, p)$: In the l^{th} iteration, this represents the mean of the LLR message passed from CN c_i in the p^{th} protograph to VN v_j in one of the $\{1, \dots, p\}$ protographs.
- $L_{\text{app},t}^{\text{max}}(j, p)$: The a posteriori mean LLR for VN v_j in the p^{th} protograph after l_{max} decoding iterations.

The DE recursion is initialized as:

$$\mu_0(j) = L_{v,t}^0(i, j, p) = \frac{2\alpha_f^2}{\sigma_n^2} = 4\gamma_f, \quad (12)$$

for all $0 < p \leq t$.

The DE equations for the RP-anytime code ensemble are presented in (13), (14), and (15), shown at the bottom of the page. The function $\phi(u)$ is defined as:

$$\phi(u) = \begin{cases} 1 - \frac{1}{\sqrt{4\pi u}} \int_{-\infty}^{\infty} \tanh\left(\frac{f}{2}\right) e^{-\frac{(f-u)^2}{4u}} df, & \text{if } u > 0 \\ 1, & \text{if } u = 0 \end{cases},$$

and the function $J(u)$ is defined as:

$$J(u) = 1 - \frac{1}{\sqrt{4\pi u}} \int_{-\infty}^{\infty} e^{-\frac{(f-u)^2}{4u}} \log_2(1 + e^{-f}) df.$$

Therefore, after $\mathbf{y}_{[1,t]}$ have been received, the BER $P_e(j, p)$ for the j^{th} variable node in the p^{th} protograph is given by:

$$P_e(j, p) = Q\left(\sqrt{\frac{L_{\text{app},t}^{\text{max}}(j, p)}{2}}\right), \quad (16)$$

where $Q(\cdot)$ is the Gaussian Q -function.

C. Asymptotic Decoding Performance Analysis

Using the derived DE equations, we present the numerical results for the proposed RP-anytime code structure. For comparison, we also include the DE-based performance of representative prior-art root-protograph-based LDPC codes and conventional anytime codes designed for BEC or AWGN channels.

In Fig. 4, we compare the asymptotic BER performance of root-protograph-based LDPC codes with that of the proposed RP-anytime codes. For the case of code rate $R=1/2$ and block fading parameter $F=2$, we consider an irregular root-protograph-based LDPC code designed in [18], with the base matrix:

$$\mathbf{B}_{\text{IRP1}, R=1/2} = \begin{bmatrix} 1 & 0 & 2 & 2 \\ 3 & 3 & 1 & 0 \end{bmatrix}.$$

For the RP-anytime code, we use the same base protograph $\mathbf{B}_{\text{IRP1}, R=1/2}$ to construct the entire code ensemble via exponential spatial coupling. In Fig. 4, we evaluate the BER of the information symbols of the IRP1-LDPC code, and the BER of the information symbols in $\mathbf{x}_1$ for the RP-anytime code, as a function of decoding delay d (i.e., decoding window length).

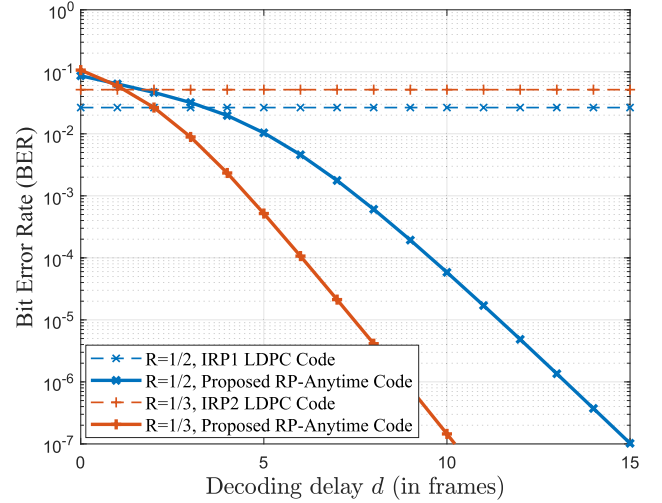


Fig. 4. Asymptotic decoding performance comparison between root-protograph-based LDPC codes and the proposed RP-anytime codes. For the $R=1/2$ case, codes operate over a non-ergodic BF channel with $F=2$ and $\gamma = (0.4, 1)$. For the $R=1/3$ case, the BF channel has $F=3$ and $\gamma = (0.1, 0.4, 0.9)$.

Additionally, we include results for codes with rate $R=1/3$ operating over block fading channels with $F=3$. The root-protograph-based LDPC code used here is the one from [29], with base matrix:

$$\mathbf{B}_{\text{IRP2}, R=1/3} = \begin{bmatrix} 1 & 0 & 0 & 2 & 0 & 3 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 3 & 2 \\ 0 & 3 & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 2 & 0 & 3 \\ 2 & 0 & 3 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 2 & 1 & 0 & 0 \end{bmatrix}.$$

We construct the corresponding RP-anytime code ensemble using this matrix and again target the BER of $\mathbf{x}_1$.

As illustrated in Fig. 4, the root-protograph-based LDPC codes do not exhibit the anytime-reliable property—their BER initially drops but does not improve with increasing d . In contrast, the proposed RP-anytime codes exhibit a consistently decreasing BER with respect to d , confirming their anytime reliability. Furthermore, the figure shows that lower code rates result in improved BER performance for the RP-anytime codes, demonstrating their robustness across a range

$$L_{c,t}^l(i, j, p) = \phi^{-1} \left[1 - \prod_{s \neq j} \left(1 - \phi \left(J^{-1} \left(\sum_{h=1}^w \Pr(h) J(L_{v,t}^l(i, s, p-h)) \right) \right) \right)^{b_{i,s}} \right] \left(\left(1 - \phi \left(J^{-1} \left(\sum_{h=1}^w \Pr(h) J(L_{v,t}^l(i, j, p-h)) \right) \right) \right)^{b_{i,j-1}} \right), \quad (13)$$

$$L_{v,t}^l(i, j, p) = \mu_0 + \sum_{s \neq i}^{b_c} b_{s,j} J^{-1} \left(\sum_{h=1}^w \Pr(h) J(L_{c,t}^{l-1}(s, j, p+h)) \right) + (b_{i,j} - 1) J^{-1} \left(\sum_{h=1}^w \Pr(h) J(L_{c,t}^{l-1}(i, j, p+h)) \right). \quad (14)$$

$$L_{\text{app},t}^{\text{max}}(j, p) = \mu_0 + \sum_{i=1}^{b_c} b_{i,j} J^{-1} \left(\sum_{h=1}^w \Pr(h) J(L_{c,t}^{l-1}(i, j, p+h)) \right). \quad (15)$$

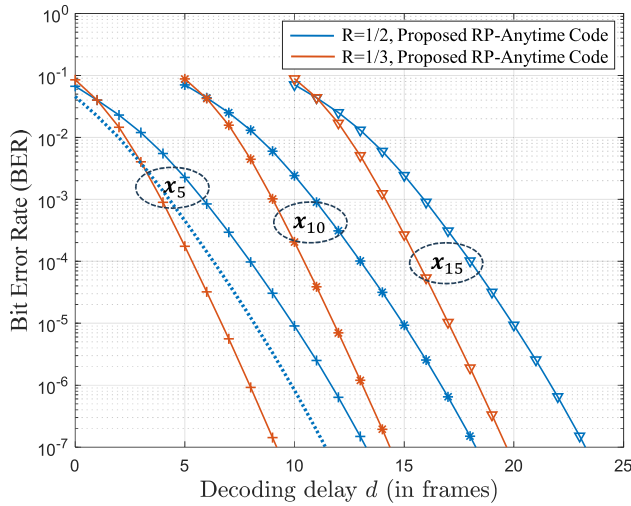


Fig. 5. Asymptotic decoding performance of different encoded frames in the proposed RP-anytime codes. For the $R=1/2$, $F=2$ case, the solid lines correspond to $\gamma = (0.5, 1)$, while the dotted line corresponds to $\gamma = (0.6, 1.2)$. For the $R=1/3$, $F=3$ case, $\gamma = (0.1, 0.5, 1)$.

of coding rates. In contrast, for the root-protograph-based LDPC codes, under the given γ configurations, the $R=1/3$ code performs even worse than the $R=1/2$ code, highlighting their sensitivity to both rate and fading conditions in anytime-coded transmissions.

Then in Fig. 5, we illustrate the asymptotic BER performance of the encoded frames x_5 , x_{10} , and x_{15} from the proposed RP-anytime codes constructed using $\mathbf{B}_{\text{IRP1}, R=1/2}$ and $\mathbf{B}_{\text{IRP2}, R=1/3}$, respectively, plotted against the decoding delay d for x_5 .

First, by comparing the BER curves for different encoded frames of the same RP-anytime code, we observe that the BER performance of each frame is a time-shifted version of the others. This indicates that the BER of a given frame depends solely on its decoding delay d and not on the frame index. Therefore, the BER performance of any single frame is representative of all frames in the RP-anytime code ensemble.

Second, we observe that the decay rate of the BER curves—i.e., the delay-exponent ε —is not constant but varies with d . This implies that, unlike anytime codes designed for BEC or AWGN channels [3], [8], [9], [10], [11], a single delay-exponent value cannot characterize the performance of RP-anytime codes over non-ergodic BF channels. Furthermore, different realizations of the received SNR vector γ also affect the BER even for the same RP-anytime code, as shown by the performance gap between the dotted and solid curves for $R=1/2$, frame x_5 .

To account for these variations and establish a more comprehensive performance metric for anytime-coded systems in non-ergodic BF channels, we introduce the *average asymptotic BER*, denoted as $\bar{A}_{\text{BER}}$. This metric represents the average BER of an encoded frame, integrated over a specified range of d and all possible realizations of the received SNR vector γ :

$$\bar{A}_{\text{BER}} = \int_{\gamma} \int_0^{d_{\max}} P_e(\mathbf{x}, d, \gamma) f(\gamma) dd d\gamma, \quad (17)$$

Algorithm 1 Anytime SNR Threshold Algorithm

Input: $\mathbf{B}$, j^{info} , λ , w , d , $\gamma_{\min}$, $\gamma_{\max}$, $l_{\max}$

```

1: for  $\gamma_1 = \gamma_{1,\min}$  to  $\gamma_{1,\max}$  step 0.001 do
2:   for  $\gamma_2 = \gamma_{2,\min}$  to  $\gamma_{2,\max}$  step 0.001 do
3:     ...
4:     for  $\gamma_F = \gamma_{F,\min}$  to  $\gamma_{F,\max}$  step 0.001 do
5:       Initialize DE recursion using (12)
6:       for  $l = 1$  to  $l_{\max}$  do
7:         Perform DE updates using (13) and (14) for
8:            $p = 1$  to  $d$ 
9:       end for
10:      Compute frame error rate for  $\mathbf{x}_1$ :
11:       $P_{\text{fe}}(1) = \sum_{j \in j^{\text{info}}} P_e(j, 1)$  using (15), (16)
12:      if  $P_{\text{fe}}(1) = 0$  then
13:        Record  $(\gamma_1^{\text{th}}, \dots, \gamma_F^{\text{th}})$  as an SNR-threshold
14:        pair
15:      end if
16:    end for
17:  end for
18: end for

```

Output: Set of SNR-threshold vectors γ^{th}

where $P_e(\mathbf{x}, d, \gamma)$ denotes the conditional average BER of frame $\mathbf{x}$ after decoding delay d under fading realization γ , and $f(\gamma)$ is the probability density function (PDF) of γ . In practice, the integration is computed over bounded intervals for both d and γ .

D. Anytime Outage Region

In addition to the anytime-reliable property, the performance of anytime codes over non-ergodic BF channels must also be evaluated in terms of their *outage region*. This concept, previously explored in [17] and [18] for root-LDPC codes and root-protograph-based LDPC codes, respectively, involves the derivation of an SNR/fading threshold. The resulting outage region serves as an analytical tool for characterizing the achievable diversity order and predicting the asymptotic codeword error behavior under various fading conditions.

In this subsection, we extend this concept by modifying the threshold derivation process to formulate an *anytime SNR/fading threshold*, which estimates the FER for each individual frame in the proposed RP-anytime code structure. Unlike traditional block codes, anytime codes continuously generate frames over time, leading to codewords of theoretically unbounded length. Consequently, estimating the WER of the entire codeword is not meaningful in this context. Instead, we focus on the reliability of each transmitted frame, as discussed in subsection III-C.

The modified anytime SNR-threshold algorithm is presented in Algorithm 1, which builds upon the DE recursions outlined in subsection III-B. The required inputs to the algorithm include: the root-protograph base matrix $\mathbf{B}$ defining the RP-anytime code ensemble; the indices of the information VNs in $\mathbf{B}$, denoted by j^{info} ; the exponential coupling rate λ ; a sufficiently large coupling width w ; the target decoding delay d ; and the range of each element in the received SNR vector $\gamma \in [\gamma_{\min}, \gamma_{\max}]$, where $\gamma_{\min} = (\gamma_{1,\min}, \dots, \gamma_{F,\min})$ and $\gamma_{\max} = (\gamma_{1,\max}, \dots, \gamma_{F,\max})$. The algorithm also requires the maximum number of decoding iterations $l_{\max}$ as input.

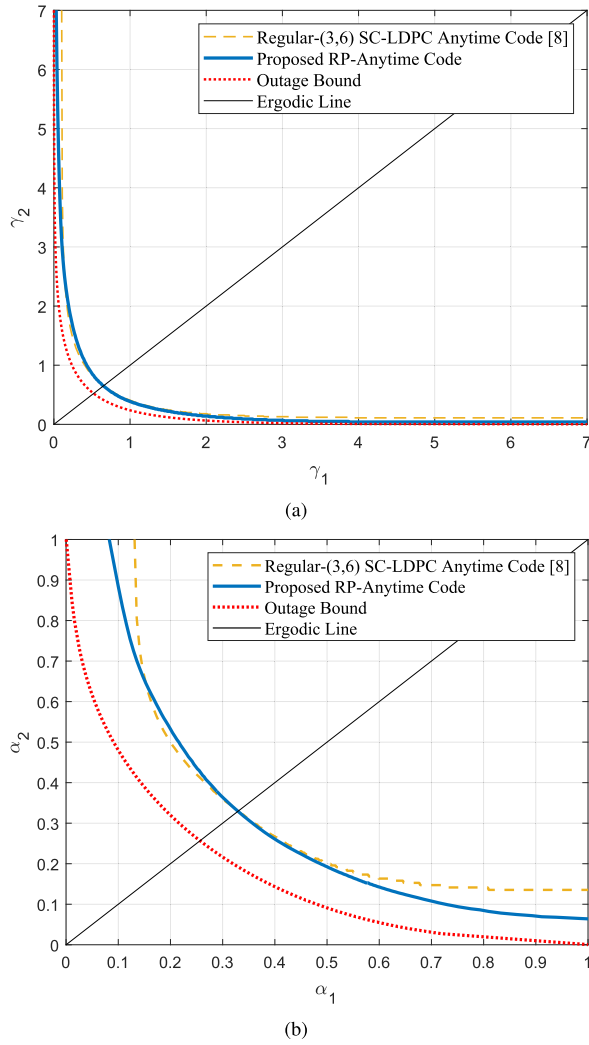


Fig. 6. (a) The anytime SNR outage boundaries; (b) the anytime fading outage boundaries, of the SC-LDPC anytime codes [8] and the RP-anytime code constructed from $\mathbf{B}_{\text{IRP1}, R=1/2}$. The parameters used are $R = 1/2$, $F = 2$, and $E_b/N_0 = 12\text{dB}$.

As discussed in the previous section, the asymptotic performance of each frame in an RP-anytime code is statistically independent of the others. Therefore, *Algorithm 1* is constructed by focusing on the error probability P_e of frame $\mathbf{x}_1$, which is representative of the performance of all other frames in the ensemble.

Once the SNR-threshold pairs γ^{th} are obtained, the corresponding fading-threshold pairs α^{th} can be computed using (3) for a given E_b/N_0 . Following the definitions in [27], the line formed by the SNR-threshold pairs is referred to as the *SNR outage boundary*, while the corresponding line formed by the fading thresholds is termed the *fading outage boundary*.

Definition: Similar to the SNR and fading outage regions defined in [18], we define the *anytime SNR and fading outage regions* as:

$$\begin{aligned} \bar{\mathcal{D}}_{\text{SNR}} &= \{(\gamma_1^{\text{th}}, \dots, \gamma_F^{\text{th}}) \in \mathbb{R}_+^F \mid P_{\text{blk}}(1) > 0\}, \\ \bar{\mathcal{D}}_{\text{Fading}} &= \{(\alpha_1^{\text{th}}, \dots, \alpha_F^{\text{th}}) \in \mathbb{R}_+^F \mid P_{\text{blk}}(1) > 0\}. \end{aligned}$$

In Fig. 6, we present both the anytime SNR outage boundaries and the anytime fading outage boundaries for two code

ensembles: the regular-(3,6) SC-LDPC anytime code [8] and the proposed RP-anytime code constructed from $\mathbf{B}_{\text{IRP1}, R=1/2}$. Both codes operate at a rate of $R=1/2$ over a non-ergodic BF channel with $F=2$ and $E_b/N_0 = 12\text{ dB}$. In the same figure, we also plot the theoretical outage bound and the ergodic line defined by $\gamma_1 = \gamma_2$ (or equivalently, $\alpha_1 = \alpha_2$) [27]. The intersection of each outage boundary with the ergodic line corresponds to the *ergodic SNR/fading threshold*, denoted as $\gamma_{\text{erg}}/\alpha_{\text{erg}}$, which is solely determined by the decoding threshold over an AWGN channel. For anytime codes, this threshold depends on the average degree distribution of the VNs.

As observed in Fig. 6, both the RP-anytime and SC-LDPC anytime codes share the same $\gamma_{\text{erg}}/\alpha_{\text{erg}}$ due to their identical VN degrees. However, the RP-anytime code exhibits a significantly smaller anytime outage region, indicating its superior robustness and ability to maintain reliable decoding under a broader range of fading conditions.

Furthermore, achieving full diversity in non-ergodic BF channels requires that the SNR outage boundary converges toward zero in each SNR dimension, i.e., as $\gamma_1 \rightarrow \infty$ or $\gamma_2 \rightarrow \infty$ for $F=2$; equivalently, the fading outage boundary should converge as $\alpha_1 \rightarrow \infty$ or $\alpha_2 \rightarrow \infty$ [18].

As shown in Fig. 6(b), although the fading outage boundary of the proposed RP-anytime code does not reach zero within the displayed range, it clearly exhibits a converging trend toward zero along both fading dimensions. A similar convergence trend is observed in the SNR domain in Fig. 6(a). This behavior suggests that the proposed RP-anytime code is capable of achieving full diversity. In contrast, the SC-LDPC anytime code does not exhibit such convergence, due to the absence of a root-protograph structure, and thus cannot guarantee full diversity over non-ergodic BF channels.

IV. PROTOGRAPH DESIGN FOR THE PROPOSED RP-ANYTIME CODES

As discussed in Section III-C, a smaller *average asymptotic BER* $\bar{A}_{\text{BER}}$ reflects a steeper decay in the BER of each frame with respect to decoding delay d , indicating that fewer decoding delays are required to achieve low error rates. Meanwhile, as shown in Section III-D, a smaller *anytime SNR outage region* $\bar{\mathcal{D}}_{\text{SNR}}$ ³ signifies that the code is closer to the theoretical outage bound, thus offering a higher diversity order and improved error performance in non-ergodic BF channels.

Therefore, a good RP-anytime code should have both a small $\bar{A}_{\text{BER}}$ and a small $\bar{\mathcal{D}}_{\text{SNR}}$ simultaneously. To achieve this, we employ a multi-objective optimization problem (MOP) approach to design a root-protograph that results in RP-anytime codes with minimized $\bar{A}_{\text{BER}}$ and $\bar{\mathcal{D}}_{\text{SNR}}$ values. The proposed MOP can be expressed as

$$\min_{\mathbf{B}} \{J_1 = \bar{A}_{\text{BER}}, J_2 = \bar{\mathcal{D}}_{\text{SNR}}\}$$

where J_1 and J_2 are the objective functions that need to be minimized.

A MOP can be formally defined as the problem of finding decision vectors $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)^T$ in the decision space,

³ $\bar{\mathcal{D}}_{\text{SNR}}$ is preferred due to its independence from E_b/N_0 .

Algorithm 2 Multi-Objective Differential Evolution for Root-Protograph Design**Input:** $D = b_v b_c, N_{\text{pop}}, G_{\text{max}}, S, C_r, b_{\text{max}}$

1: **Initialization:** Create an initial population $\mathcal{P}^1 = \{\mathbf{x}_1^1, \mathbf{x}_2^1, \dots, \mathbf{x}_{N_{\text{pop}}}^1\}$ by randomly generating matrices $\mathbf{x}_i^1 = \{x_{i1}^1, x_{i2}^1, \dots, x_{iD}^1\}$ that meet the predefined code constraints.

2: Compute the fitness value, i.e. $\bar{\mathcal{A}}_{\text{BER}}$ and $\bar{\mathcal{D}}_{\text{SNR}}$, for each matrix in $\mathcal{P}^1$.

3: **for** $g = 1 : G_{\text{max}}$ **do**

4: **Mutation:** Generate a mutant vector $\mathbf{m}_i^{g+1}, i \in \{1, \dots, N_{\text{pop}}\}$:

$$\mathbf{m}_i^{g+1} = \mathbf{x}_{r_1}^g + S \cdot (\mathbf{x}_{r_2}^g - \mathbf{x}_{r_3}^g),$$

5: where $\mathbf{x}_{r_1}^g, \mathbf{x}_{r_2}^g$, and $\mathbf{x}_{r_3}^g$ are randomly chosen from $\mathcal{P}^g$.

6: **Crossover:** Generate a trial vector $\mathbf{v}_i^{g+1}, i \in \{1, \dots, N_{\text{pop}}\}$ as:

$$v_{ij}^{g+1} = \begin{cases} m_{ij}^{g+1}, & \text{with probability } C_r \\ x_{ij}^{g+1}, & \text{with probability } 1 - C_r \end{cases}$$

7: where $j = 1, 2, \dots, D$. If the generated vector $\mathbf{v}_i^{g+1}$ does not adhere to the code constraints, repeat this crossover process for $\mathbf{v}_i^{g+1}$.

8: **Selection:** Perform the selection step for $i \in \{1, \dots, N_{\text{pop}}\}$:

9: (i) If the trial vector $\mathbf{v}_i^{g+1}$ dominates the current solution $\mathbf{x}_i^g$, then update $\mathbf{x}_i^{g+1}$ to be $\mathbf{v}_i^{g+1}$;

10: (ii) If the current solution $\mathbf{x}_i^g$ dominates the trial vector $\mathbf{v}_i^{g+1}$, then retain $\mathbf{x}_i^g$ as $\mathbf{x}_i^{g+1}$;

11: (iii) If neither $\mathbf{x}_i^g$ nor $\mathbf{v}_i^{g+1}$ dominates the other, include $\mathbf{v}_i^{g+1}$ in the population.

12: After selection, the population size will range from N_{pop} to $2N_{\text{pop}}$. Filter out the best N_{pop} individuals to form $\mathcal{P}^{g+1}$.

13: **end for**

Output: $\mathcal{P}^{G_{\text{max}}+1}$

which satisfy all constraints and optimize the objective function vector $\min\{f_1(\mathbf{X}), f_2(\mathbf{X}), \dots, f_k(\mathbf{X})\}$. The constraints define the feasible region, and any vector $\mathbf{X}$ within this feasible region is called a feasible solution [31].

In MOPs, no single global solution exists. Therefore, the concept of Pareto dominance is introduced [32]. A feasible solution $\mathbf{X}_1 = (x_{1,1}, x_{1,2}, \dots, x_{1,n})^T$ is said to dominate another feasible solution $\mathbf{X}_2 = (x_{2,1}, x_{2,2}, \dots, x_{2,n})^T$ if both of the following conditions are satisfied:

$$\begin{aligned} f_i(\mathbf{X}_1) &\leq f_i(\mathbf{X}_2), \quad \forall i \in \{1, 2, \dots, k\}, \\ f_i(\mathbf{X}_1) &< f_i(\mathbf{X}_2), \quad \exists i \in \{1, 2, \dots, k\}. \end{aligned}$$

In other words, a solution is said to dominate another if it is strictly better in at least one objective and no worse in any of the others.

In this work, we aim to identify a set of Pareto-dominant root-protograph base matrices $\mathbf{B}$ that simultaneously minimize $\bar{\mathcal{A}}_{\text{BER}}$ and $\bar{\mathcal{D}}_{\text{SNR}}$. To solve this multi-objective optimization problem, we adopt the multi-objective differential evolution (MODE) algorithm introduced in [33] and [34], which is well suited for exploring complex, non-convex search spaces.

Algorithm 2 summarizes the MODE-based optimization procedure employed in this work. Each individual $\mathbf{x}$ in the

population represents a candidate root-protograph base matrix $\mathbf{B}$, expressed in an array form and subject to a set of predefined structural constraints. The inputs to the algorithm include the base-matrix dimension D , the population size N_{pop} , the maximum number of generations G_{max} , the mutation scaling factor $S \in [0, 1]$, the crossover probability $C_r \in [0, 1]$, and the maximum entry value b_{max} .

In Algorithm 2, $\mathcal{P}^g = \{\mathbf{x}_1^g, \mathbf{x}_2^g, \dots, \mathbf{x}_{N_{\text{pop}}}^g\}$ denotes the population at the g -th generation, where each individual $\mathbf{x}_i^g$ represents a feasible root-protograph base matrix $\mathbf{B}$. During the mutation step, a mutant vector $\mathbf{m}_i^{g+1}$ is generated by combining three randomly selected parent individuals from $\mathcal{P}^g$. A trial vector $\mathbf{v}_i^{g+1}$ is then obtained through crossover between $\mathbf{m}_i^{g+1}$ and the current individual $\mathbf{x}_i^g$, and subsequently evaluated and selected based on Pareto dominance.

The algorithm begins by randomly generating N_{pop} candidate base matrices $\mathbf{B}$ (denoted as $\mathbf{x}$), each of which must satisfy the following constraints:

- The base matrix $\mathbf{B}$ must preserve the predefined root structure. Specifically, for each information VN type associated with the fading gain α_f , the corresponding columns of $\mathbf{B}$ must contain at least one nonzero entry connecting it to a type- f root-check (i.e., a type- f CN). This constraint guarantees that the full-diversity property is maintained.
- Each entry $b_{i,j}$ of $\mathbf{B}$ is restricted to a non-negative integer bounded by b_{max} , i.e., $b_{i,j} \in \{0, 1, \dots, b_{\text{max}}\}$. Here, $b_{i,j}$ represents the number of parallel edges between the corresponding VN and CN types. The bound b_{max} limits node degrees and decoding complexity, and mitigates the formation of short cycles after lifting.
- The degree of each CN, given by the sum of the entries in the corresponding row of $\mathbf{B}$, must be at least 2. This constraint excludes degree-one CNs and ensures stable iterative decoding behavior.

At each generation, the MODE procedure evolves the population through mutation, crossover, and selection. Mutation is performed by combining three randomly selected parent matrices according to the standard differential evolution rule, producing a mutant matrix that introduces controlled structural variations. A crossover operation is then applied to generate a trial matrix by mixing the mutant and the current solution with probability C_r . If the generated trial matrix violates any of the structural constraints, the crossover step is repeated until a feasible candidate is obtained.

The fitness of each candidate base matrix is evaluated using the two objective metrics $\bar{\mathcal{A}}_{\text{BER}}$ and $\bar{\mathcal{D}}_{\text{SNR}}$, which are computed after constructing the candidate into a valid RP-anytime code following the rules in Section III-A with the prescribed exponential parameter λ , thereby ensuring that the anytime property is preserved during fitness evaluation. Selection is carried out based on Pareto dominance, where fast non-dominated sorting and crowding-distance techniques [35] are employed to retain the best N_{pop} individuals for the next generation. The evolutionary process continues until the maximum generation index G_{max} is reached.

TABLE I

$\bar{A}_{\text{BER}}$ AND $\bar{D}_{\text{SNR}}$ OF ANYTIME CODES CONSTRUCTED FROM DIFFERENT PROTOGRAPHS WITH $R = 1/2$ AND $R = 1/3$

Code rate	Protograph base matrix	$\bar{A}_{\text{BER}}$	$\bar{D}_{\text{SNR}}$
$R = 1/2$	$\mathbf{B}_{\text{Reg36}}$ [30]	13.973	1.729
	$\mathbf{B}_{\text{IRP1}}$ [18]	8.909	1.073
	$\mathbf{B}_{\text{D1}}$	6.188	0.784
$R = 1/3$	$\mathbf{B}_{\text{Reg46}}$ [30]	10.535	1.343
	$\mathbf{B}_{\text{IRP2}}$ [29]	6.253	0.884
	$\mathbf{B}_{\text{D2}}$	3.014	0.476

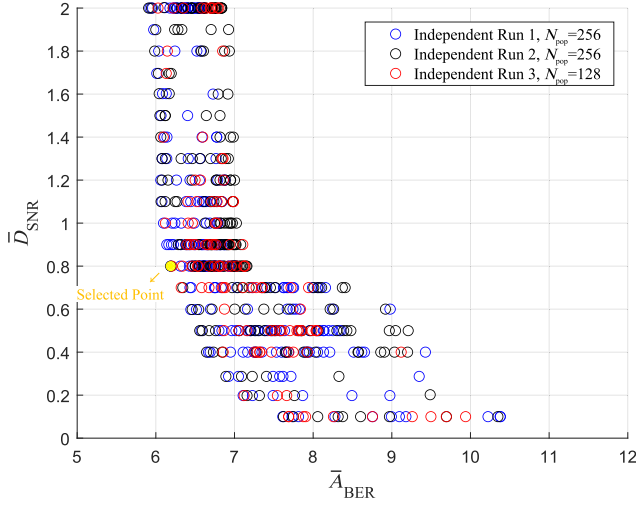


Fig. 7. Pareto fronts obtained from independent MODE runs with different population sizes. The marked point indicates the selected protograph.

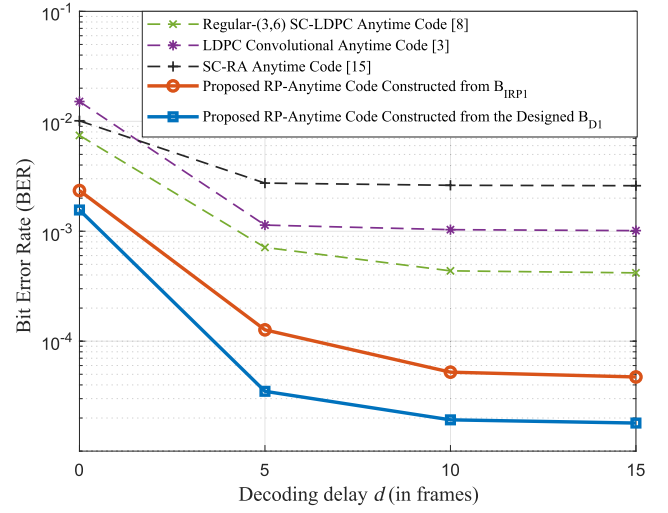
V. NUMERICAL RESULTS

The construction parameters of RP-anytime codes, such as λ , coupling width w , decoding delay d , and lifting factor M_f , can be adjusted according to application-specific latency and reliability requirements. In this work, these parameters are fixed to representative values to enable consistent performance comparison. The impact of varying these parameters is further investigated in the Appendix.

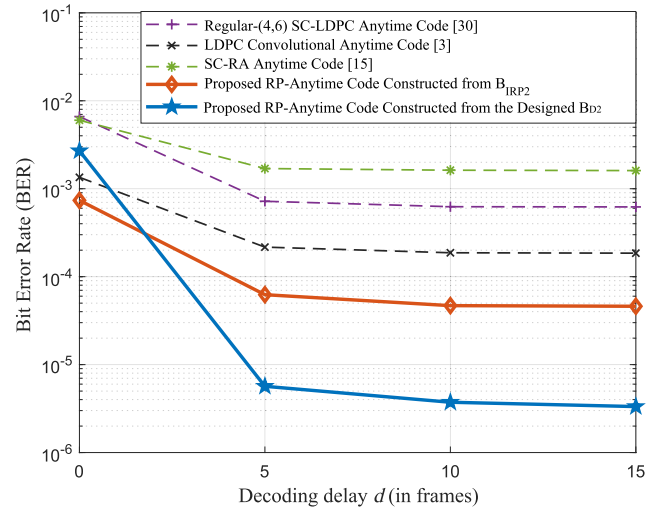
Using *Algorithm 2*, we design root-protographs with minimized $\bar{A}_{\text{BER}}$ and $\bar{D}_{\text{SNR}}$ for dimensions $D = 4 \times 8$ with code rate $R = 1/2$, and for dimensions $D = 6 \times 9$ with code rate $R = 1/3$, respectively. The input parameters are set as follows: $N_{\text{pop}} = 8 \times D$, $G_{\text{max}} = 60$, $S = 0.4$, $C_r = 0.7$, and $b_{\text{max}} = 3$. The construction parameter λ is fixed as $\lambda = 0.5$, and the objective metrics are computed over the predefined delay range $d \in [0, 25]$, covering the operating region considered in this work.

During the design process, to reduce the computational complexity of $\bar{A}_{\text{BER}}$ and $\bar{D}_{\text{SNR}}$ for each individual, we select 11 points to compute $\bar{A}_{\text{BER}}$ and 3 representative points on the anytime SNR outage boundary to compute $\bar{D}_{\text{SNR}}$. The base matrices of the two designed root-protographs are:

$$\mathbf{B}_{\text{D1}, R=1/2} = \begin{bmatrix} 1 & 0 & 0 & 0 & 3 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 2 & 2 \\ 2 & 2 & 0 & 0 & 1 & 0 & 0 & 0 \\ 2 & 0 & 1 & 2 & 0 & 1 & 0 & 0 \end{bmatrix},$$



(a)



(b)

Fig. 8. (a) BER of m_5 for anytime codes with $R = 1/2$ over non-ergodic BF channels with $F = 2$ and $E_b/N_0 = 18$ dB. (b) BER of m_5 for anytime codes with $R = 1/3$ over non-ergodic BF channels with $F = 3$ and $E_b/N_0 = 12$ dB.

$$\mathbf{B}_{\text{D2}, R=1/3} = \begin{bmatrix} 1 & 0 & 0 & 3 & 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 \\ 2 & 0 & 1 & 1 & 0 & 0 & 3 & 0 & 0 \\ 1 & 2 & 3 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix}.$$

In Table I, for codes with rate $R = 1/2$ we present the average asymptotic BER $\bar{A}_{\text{BER}}$ and the anytime SNR outage region $\bar{D}_{\text{SNR}}$ values for the P-anytime codes constructed from $\mathbf{B}_{\text{Reg36}}$, $\mathbf{B}_{\text{IRP1}}$, as well as our proposed $\mathbf{B}_{\text{D1}, R=1/2}$. For code with rate $R = 1/3$, we present the average asymptotic BER $\bar{A}_{\text{BER}}$ and the anytime SNR outage region $\bar{D}_{\text{SNR}}$ values for the P-anytime codes constructed from $\mathbf{B}_{\text{Reg46}}$, $\mathbf{B}_{\text{IRP2}}$, as well as our proposed $\mathbf{B}_{\text{D2}, R=1/3}$. According to these tables, the designed $\mathbf{B}_{\text{D1}, R=1/2}$ and $\mathbf{B}_{\text{D2}, R=1/3}$ results in the smallest $\bar{A}_{\text{BER}}$ and $\bar{D}_{\text{SNR}}$ values among all protographs, while $\mathbf{B}_{\text{Reg36}}$ and $\mathbf{B}_{\text{Reg46}}$ yields the largest values, due to their lack of root-protograph structure.

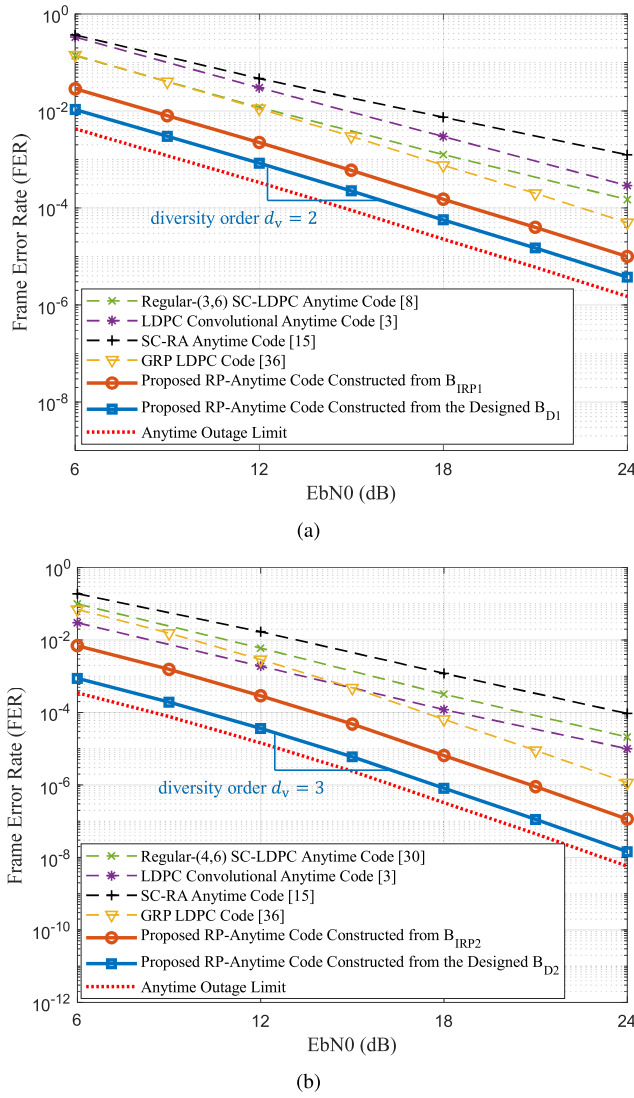


Fig. 9. FER performance of various codes with comparable codeword lengths (approximately 1000 bits) over non-ergodic BF channels with (a) $F = 2$ and (b) $F = 3$. For the anytime codes, this is achieved using a decoding delay of 25 frames, with a frame length of 40 bits for $R = 1/2$ and 45 bits for $R = 1/3$, respectively.

To demonstrate the robustness of the proposed MODE-based optimization, Fig. 7 illustrates the Pareto fronts obtained from multiple independent runs under identical hyperparameter settings. Two independent runs with population size $N_{\text{pop}} = 256$ produce nearly overlapping Pareto fronts, indicating stable convergence behavior and low sensitivity to random initialization. In addition, a run with a reduced population size ($N_{\text{pop}} = 128$) yields a slightly sparser front while preserving a similar overall trade-off geometry. These observations confirm the stability and robustness of the optimization process.

To evaluate the computational performance, both MODE and random search were run on a Lenovo laptop equipped with an AMD Ryzen 7 PRO 3700U CPU and 16 GB RAM, using 8 MATLAB parallel workers. The proposed MODE with $N_{\text{pop}} = 256$ required an average runtime of approximately 6618 seconds to obtain the Pareto front shown in Fig. 7. In comparison, random search required approximately 48830 seconds to obtain a similar Pareto front, which is about

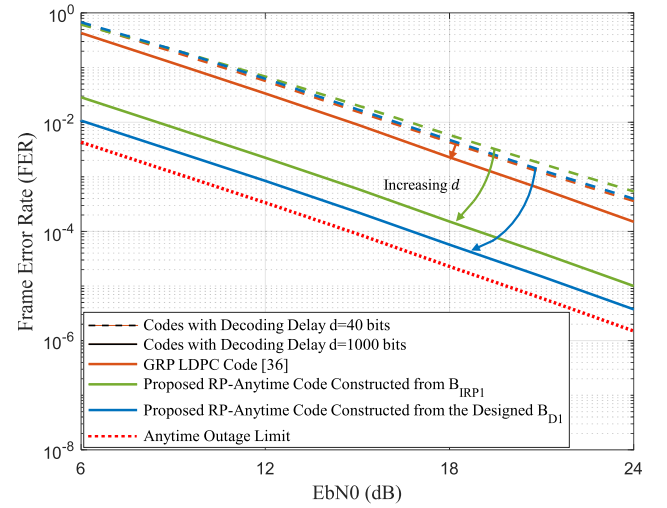


Fig. 10. FER performance of various codes over non-ergodic BF channels with $F = 2$. The dashed and solid lines correspond to decoding delays of 40 and 1000 bits, respectively, which translate into codeword lengths of 40 and 1000 bits for the GRP-LDPC code [36], and decoding delays of 1 and 25 frames with a frame length of 40 bits for the anytime codes.

7.38 times longer than that of MODE. This result demonstrates the computational efficiency and practical feasibility of the proposed MODE-based design method.

The final selected protograph (i.e., $\mathbf{B}_{D1, R=1/2}$) corresponds to a balanced trade-off point on the Pareto front, achieving simultaneously low $\bar{\mathcal{A}}_{\text{BER}}$ and favorable $\bar{\mathcal{D}}_{\text{SNR}}$ performance.

Fig. 8 compares the anytime properties of our proposed RP-anytime codes with prior-art anytime codes in non-ergodic BF channels. In Fig. 8(a), we show the BER performance of frame m_5 versus the decoding delay d for regular-(3,6) SC-LDPC anytime code [8], LDPC convolutional anytime code [3], the SC-RA anytime code [15], the proposed RP-anytime codes constructed from $\mathbf{B}_{\text{IRP1}}$ and the designed $\mathbf{B}_{D1, R=1/2}$. All codes have a code rate of $R = 1/2$, a frame length of 40 bits, and are transmitted over a non-ergodic BF channel with $F = 2$ at $E_b/N_0 = 18$ dB. Comparing the two thick solid lines, which are our proposed RP-anytime code structure, we can tell that this proposed structure provides huge improvement compared with prior-art anytime codes without RP structure, demonstrating the effectiveness of the proposed RP-anytime structure. Furthermore, if comparing the two proposed RP-anytime codes, the one that constructed from $\mathbf{B}_{D1}$ which is designed from the MODE algorithm shows further improvement in BER, proving the effectiveness of the modified MODE algorithm design.

Similarly, Fig. 8(b) shows the BER performance of various anytime codes with a code rate of $R = 1/3$ and a frame length of 45 bits over non-ergodic BF channels with $F = 3$ at $E_b/N_0 = 12$ dB. This setting is intended to better illustrate the performance differences. It can be expected that with $E_b/N_0 = 18$ dB, the overall BER performance would be significantly lower than that at 12 dB. The codes involved in Fig. 8(b) are the regular-(4,6) SC-LDPC anytime code [30], the LDPC convolutional anytime code [3], the SC-RA anytime code [15], the proposed RP-anytime codes constructed from $\mathbf{B}_{\text{IRP2}}$ and the designed $\mathbf{B}_{D2, R=1/3}$.

TABLE II
SUMMARY OF SIMULATION SETTINGS FOR NUMERICAL RESULTS (FIGS. 8–10)

Parameter	$R = 1/2$	$R = 1/3$
Applicable figures	Fig. 8(a), Fig. 9(a), Fig. 10	Fig. 8(b), Fig. 9(b)
Performance metric	Fig. 8: BER vs. delay Figs. 9–10: FER vs. E_b/N_0	Fig. 8: BER vs. delay Fig. 9: FER vs. E_b/N_0
Frame length N (bits)	40	45
Decoding delay d (frames)	Fig. 8: 0–15 Figs. 9–10: 25	Fig. 8: 0–15 Fig. 9: 25
Codeword length (bits)	Fig. 8: 0–600 Figs. 9–10: 1000	Fig. 8: 0–675 Fig. 9: 1125
Max. decoding iterations $l_{\max}$	100	100
Lifting factor M_f	$\mathbf{B}_{\text{IRP1}} : 10$ $\mathbf{B}_{\text{D1}} : 5$	$\mathbf{B}_{\text{IRP2}} : 5$ $\mathbf{B}_{\text{D2}} : 5$
Coupling width w	25	25
Coupling parameter λ	0.5	0.5
Fading blocks F	2	3
Random seeds	Independent random realizations of code construction and channel	

In Fig. 9, we illustrate the diversity-order properties of the proposed codes by comparing their FER performance with that of various prior-art anytime codes and root-protograph LDPC codes, plotted as a function of E_b/N_0 in dB. Fig. 9(a) presents results for codes with rate $R = 1/2$ over non-ergodic BF channels with $F = 2$, while Fig. 9(b) shows the case for $R = 1/3$ over BF channels with $F = 3$. All codes are configured to have comparable codeword lengths (approximately 1000 bits). For the anytime codes, this is achieved by using a decoding delay of 25 frames, with a frame length of 40 bits for $R = 1/2$ and 45 bits for $R = 1/3$, respectively.

In Fig. 9(a), three codes achieve full diversity with diversity order $d_v = 2$: the GRP-LDPC code [36], and the proposed RP-anytime codes constructed from $\mathbf{B}_{\text{IRP1}}$ and from the designed protograph $\mathbf{B}_{\text{D1}}$. Although all three codes attain full diversity, the proposed RP-anytime code structure provides notably better FER performance, approaching the theoretical outage limit. In particular, the code constructed from the designed protograph $\mathbf{B}_{\text{D1}}$ achieves the closest match to the outage limit, thereby further confirming the effectiveness of the proposed *Algorithm 2*. In contrast, prior-art anytime codes perform poorly under non-ergodic BF conditions. The SC-RA anytime code [15] exhibits the worst FER with a diversity order of $d_v \approx 1.025$, followed by the LDPC convolutional anytime code [3] with $d_v \approx 1.291$, and the regular-(3,6) SC-LDPC anytime code [8] with $d_v \approx 1.251$.

Similar trends are observed in Fig. 9(b), where full diversity is defined as $d_v = 3$. Again, the proposed RP-anytime codes constructed from either $\mathbf{B}_{\text{IRP2}}$ or the designed $\mathbf{B}_{\text{D2}}$ achieve both full diversity and the best FER performance among all evaluated codes.

Furthermore, in Fig. 10, we separately analyze the three full-diversity codes. The dashed lines indicate a decoding delay of $d = 40$ bits, which corresponds to a codeword length of

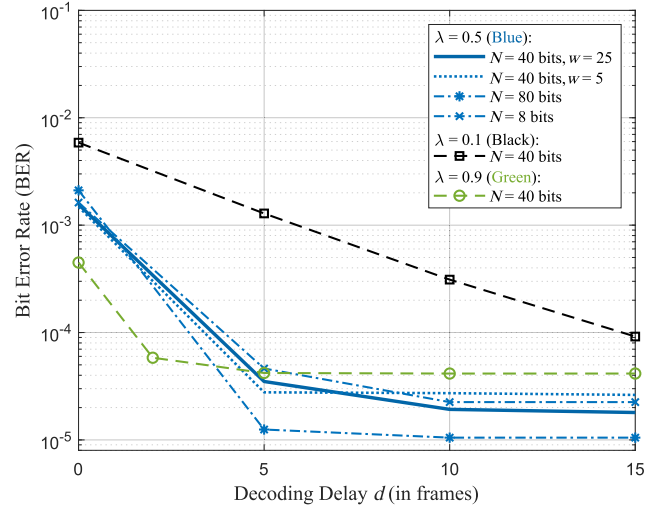


Fig. 11. BER of $\mathbf{m}_5$ versus decoding delay for different parameter settings of RP-anytime codes constructed from $\mathbf{B}_{\text{D1}, R=1/2}$ over non-ergodic BF channels with $F = 2$ at $E_b/N_0 = 18$ dB.

40 bits for the GRP-LDPC code [36], and to a decoding delay of one frame with a frame length of 40 bits for the anytime codes. The solid lines represent $d = 1000$ bits, corresponding to a codeword length of 1000 bits for the GRP-LDPC code [36], and to a decoding delay of 25 frames with the same frame length of 40 bits for the anytime codes. As observed, increasing the decoding delay significantly improves the FER performance of the proposed RP-anytime codes, bringing them much closer to the theoretical outage limit. In contrast, the GRP-LDPC code [36] exhibits only limited performance improvement with increasing decoding delay. This observation highlights the fundamental advantage of the anytime coding structure, i.e., its ability to achieve progressively improved reliability as the decoding delay increases.

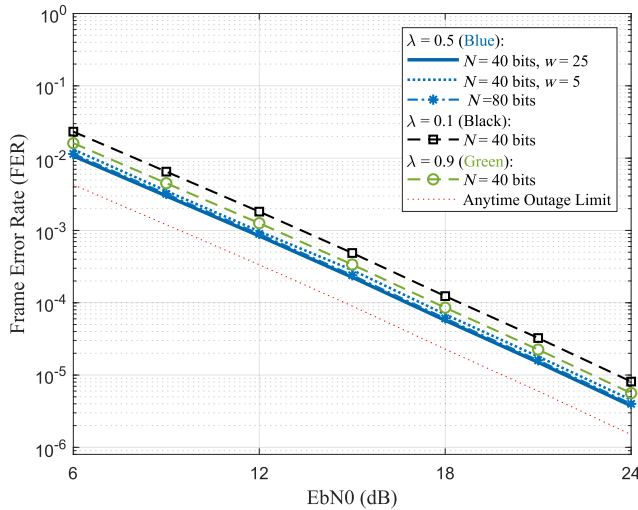


Fig. 12. FER performance for different parameter settings of RP-anytime codes constructed from $\mathbf{B}_{D1,R=1/2}$ over non-ergodic BF channels with $F = 2$. The codeword length is 600 bits for a frame length of $N = 40$ bits (decoding delay of 15 frames), and 640 bits for a frame length of $N = 80$ bits (decoding delay of 8 frames).

For clarity and reproducibility, the detailed simulation settings corresponding to Figs. 8–10 are summarized in Table II.

VI. CONCLUSION

This paper introduces a novel RP-anytime coding framework tailored for non-ergodic BF channels, a previously unexplored setting in the context of anytime communications. By modifying the DE analysis, we define two asymptotic performance metrics—average asymptotic BER $\bar{A}_{\text{BER}}$ and anytime outage region $\bar{D}_{\text{SNR}}$ —which effectively characterize the anytime reliability and diversity of the proposed codes. Based on these metrics, we further employ a MODE algorithm (Algorithm 2) to design optimized RP-anytime codes for various code rates. Both asymptotic analysis and finite-length simulations confirm that the proposed RP-anytime codes simultaneously achieve excellent anytime reliability, full diversity, and near-outage-limit error performance. The results clearly demonstrate the superiority of the RP-anytime coding structure over existing root-protograph LDPC codes and conventional anytime codes, establishing its effectiveness and robustness for delay-sensitive applications in slowly fading wireless environments.

APPENDIX

In this appendix, we provide supplementary analysis on the influence of different parameter settings on the finite-length performance of the proposed RP-anytime codes.

Fig. 11 shows the BER of $\mathbf{m}_5$ versus decoding delay for codes constructed from $\mathbf{B}_{D1,R=1/2}$ over non-ergodic BF channels with $F = 2$ at $E_b/N_0 = 18$ dB. The blue, black, and green curves correspond to $\lambda = 0.5, 0.1$, and 0.9 , respectively. Unless otherwise specified, the coupling width is set to $w = 25$.

We first examine the impact of the coupling width under $\lambda = 0.5$. For $N = 40$ bits, reducing w from 25 to 5 slightly improves the waterfall region but results in a higher error floor. This behavior suggests that the coupling width

primarily regulates the trade-off between convergence speed and error-floor suppression, rather than uniformly improving BER performance.

Next, under $\lambda = 0.5$ and $w = 25$, the BER curves for different block lengths $N = 8, 40$, and 80 bits (with lifting factors $M_f = 1, 5$, and 10 , respectively) exhibit similar trends. Although $N = 80$ bits achieves the lowest BER, the performance gap among the block lengths remains within the same order of magnitude. This indicates that increasing the lifting factor provides limited but consistent performance improvement. The gain mainly comes from reduced short cycles and improved graph connectivity, without changing the asymptotic behavior.

When comparing different values of λ , the code with $\lambda = 0.1$ exhibits a significantly slower decay rate, indicating weaker anytime reliability. In contrast, $\lambda = 0.9$ achieves faster initial decay at small decoding delays but quickly reaches a considerably higher error floor than $\lambda = 0.5$. Overall, $\lambda = 0.5$ provides the most balanced delay–reliability trade-off under the considered setting.

It should be noted that although $\lambda = 0.5$ yields a favorable trade-off in this paper for illustration purposes, the optimal choice of λ may vary depending on specific delay and reliability requirements in different application scenarios.

Fig. 12 presents the FER performance as a function of E_b/N_0 , using the same RP-anytime code configurations as in Fig. 11. In this case, the codeword length is 600 bits for $N = 40$ bits (corresponding to a decoding delay of 15 frames) and 640 bits for $N = 80$ bits (decoding delay of 8 frames).

Among the considered configurations, the setting $\lambda = 0.5$, $N = 40$ bits, and $w = 25$ achieves the best overall FER performance. The case with $N = 80$ bits exhibits nearly identical FER behavior. This similarity can be attributed to the shorter decoding delay in the $N = 80$ configuration, which partially offsets the potential performance gain from the longer frame length. In contrast, reducing the coupling width to $w = 5$, or changing λ to 0.1 or 0.9 , leads to degraded FER performance.

One important observation is that although these parameters affect the anytime reliability and decoding error performance, all RP-anytime codes still achieve full diversity. This further confirms that the full-diversity property is inherently guaranteed by the underlying root-protograph structure used to construct the RP-anytime codes, and is not affected by the tuning of the coupling or lifting parameters.

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techniques for wireless

Xiaoxi Yu received the B.Eng. degree in electrical and electronics engineering and the M.Sc. degree in wireless communications and signal processing from the University of Bristol, Bristol, U.K., in 2016 and 2018, respectively, and the Ph.D. degree in electrical and electronic engineering from Nanyang Technological University, Singapore, in 2024. She is currently a Research Fellow with Nanyang Technological University. Her research interests include information and coding theory, LDPC/protograph codes, anytime coding, HARQ schemes, and coding



modulation, sparse code

Zhaojie Yang received the B.Sc. degree in electronic engineering from Lingnan Normal University, China, in 2016, and the Ph.D. degree in communication engineering from Guangdong University of Technology, China, in 2021. He is currently an Associate Professor with the School of Information Engineering, Guangdong University of Technology. From 2022 to 2025, he was a Research Fellow with the School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore. His research interests include channel coding, coded



campus-wide NTU-NXP

Yong Liang Guan (Senior Member, IEEE) received the Bachelor of Engineering degree (Hons.) from the National University of Singapore, and the Ph.D. degree from Imperial College London, U.K. He is currently an Associate Vice President with Nanyang Technological University (NTU), Singapore, and a Professor of communication engineering with the School of Electrical and Electronic Engineering, NTU, where he has founded and is leading the Continental-NTU Corporate Research Laboratory. He also led the successful deployment of the campus-wide NTU-NXP V2X Test Bed in NTU. He has published an invited monograph, two books, and more than 540 journal and conference papers. He has secured over S90 million of external research funding. He has more than 40 filed patents and 13 granted patents (some of which were licensed to NXP, Continental). His research interests broadly include coding and signal processing for communication systems and data storage systems. He is an Editor of IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY and a Distinguished Speaker of the IEEE Vehicular Technology Society.